

Ruan van Mazijk

Technical Coordinator

C4 EcoSolutions & TAG International

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Ecologist, evolutionary biologist,
botanist, data visualiser, writer

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Languages	English (native), Afrikaans (fluent), R (10 years)	MSc in Biological Sciences (UCT)	2023
Tools	R Markdown, $\text{\LaTeX}$, bash, git, GitHub, MS Office	Under Profs Muasya, Verboom & West	
Skills	public speaking, technical writing, copyediting, statistical modelling, data manipulation, visualisation, GIS (in R), phylogeny inference & comparative methods	BSc Hons in Biological Sciences (UCT)	2017
		With distinction	
		BSc in Ecology & Evolution, Applied Biology (UCT)	2016
		With distinction in both majors & the degree	

Publications

Van Mazijk , McGregor, Duker, Butynski, Schutgens, Schleicher, Graham, Rees, Eldabaa, Mohamed, Almalki & Lee. Arthropod communities in desert ecosystems of Alula County, Saudi Arabia. Submitted to <i>Zoology in the Middle East</i> .	2025
Ismail, McGregor, Duker, Liversage, Mills, van Mazijk , Butynski, Schutgens, Schleicher, Graham, Rees, Eldabaa, Mohamed, Almalki & Lee. Organic carbon stocks in aboveground biomass and soils in hyper-arid Alula County, Saudi Arabia. Submitted to <i>Carbon Balance and Management</i> .	2025
McGregor, Ismail, Duker, Liversage, Maharaj, Mills, van Mazijk , Butynski, Schutgens, Schleicher, Graham, Rees, Eldabaa, Mohamed, Almalki & Lee. Allometric equations for hyper-arid desert plant species of Alula County, Saudi Arabia. Submitted to <i>Carbon Balance and Management</i> .	2025
Mills & van Mazijk . Permutation tests to identify significant constraint or promotion within biological scatterplots. <i>Ecol. Evol.</i> 14. 10.1002/ece3.70584	2024
Jiménez-Mejías, Manzano, Gowda, Krell, Lin, Martín-Bravo <i>et al.</i> Protecting stable biological nomenclatural systems enables universal communication: A collective international appeal. <i>BioScience</i> . 74(7). 10.1093/biosci/biae043.	2024
Mills, van Mazijk , Allen & Strydom. Soil geochemistry & constraint of tree seedlings immediately after germination on <i>Macrotermes</i> termite mounds in the Kruger National Park, South Africa. <i>Ecol. Evol.</i> 14(5). 10.1002/ece3.11348.	2024
Van Mazijk , West, Verboom, Elliott, Bureš & Muasya. Genome size variation in Cape schoenoid sedges (Cyperaceae: Schoeneae) & its ecophysiological consequences. <i>Am. J. Bot.</i> 111(8). 10.1002/ajb2.16315	2024
Mills, Duker, Lechmere-Oertel, van Mazijk & Potts. Science is the fuel required for lifting ecosystem restoration into the orbit of hundreds of millions of hectares. <i>Front. Cons. Sci.</i> 4. 10.3389/fcosc.2023.1283262.	2023
Van Mazijk , Cramer & Verboom. Environmental heterogeneity explains contrasting plant species richness between the South African Cape & southwestern Australia. <i>J. Biogeog.</i> 48. 10.1111/jbi.14118.	2021
Elliott, van Mazijk , Barrett, Bruhl, Joly, Muthaphuli, Wilson & Muasya. Global dispersal & diversification of the genus <i>Schoenus</i> (Cyperaceae) from the Western Australian biodiversity hotspot. <i>J. Syst. Evol.</i> 59(4). 10.1111/jse.12742.	2021
Elliott, Laidler, Muasya, Muthaphuli & van Mazijk . Solving the sedge mystery. <i>Veld & Flora</i> 105(3):20–25.	2019
Cramer, Wootton, van Mazijk & Verboom. New regionally-modelled soil layers improve prediction of vegetation type relative to that based on global soil models. <i>Div. Distrib.</i> 25(11). 10.1111/ddi.12973.	2019
Van Mazijk , Smyth, Weideman & West. Isotopic tracing of stormwater in the urban Liesbeek River. <i>Water SA</i> 44(4). 10.4314/wsa.v44i4.16.	2018

Professional memberships

South Afr. Association of Botanists	2017–	Data wrangling & manipulation in R (Organiser, lead, for Dept. Biological Sciences, UCT)	2019
Southern Afr. Society for Systematic Biology	2018–	Teaching Biodiversity for High School Life Science Teachers (Speaker, for SDU ¹)	2019
↔ Council student representative	2019–2021	Scientific study design & data analysis in R (Tutor, for SEEC ² -ACCESS ³)	2018, 2019

Workshop instruction

Postgraduate

¹SDU: Schools Development Unit, UCT

²SEEC: Centre for Statistics in Ecology, the Environment & Conservation, UCT

³ACCESS: Applied Centre for Climate & Earth Systems, NRF

Teaching assistance

Angiosperm diversity	Undergraduate
Principles of ecology & evolution	2018, 2019, 2022
Scientific study design & data analysis in R	2019, 2020
Introductory statistics in R	2017–2020
	2016–2018

References

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Dr Robbert Duker robbert.duker@c4es.co.za

Dept. Biological Sciences, UCT

Prof A. Muthama Muasya muthama.muasya@uct.ac.za

Prof G. Anthony Verboom tony.verboom@uct.ac.za

Prof Adam G. West adam.west@uct.ac.za

Prof Michael D. Cramer michael.cramer@uct.ac.za

Funding & awards

UCT Postgraduate Publication Incentive Funding	2023
NRF Innovation Masters Scholarship	2019
SAAB MSc Student Bursary (inaugural recipient)	2018, 2019
UCT Once-Off Award	2018
UCT Dorothy Cameron Scholarship	2018
UCT Masters Research Scholarship	2018
NRF Innovation Honours Scholarship	2017
SAAB Honours Scholarship	2017
UCT Science Faculty Scholarship	2015, 2016

⁴GENUS: DSI-NRF Centre of Excellence in Palaeosciences

⁵IBS: International Biogeography Society

⁶iDiv: German Centre for Integrative Biodiversity Research, Halle-Jena-Leipzig

⁷CFR: Cape Floristic Region

Conferences, talks & press

SASSB XVI (3 rd Best Student Presentation)	2023	Best Systematics MSc Presentation (tied)	
SASSB & GENUS ⁴ sponsored attendance		UCT Travel Grant for attendance	
SAAB Postgraduate Virtual Symposium	2022	satRday Cape Town (Attended)	2018
MEDECOS XVII (NRF sponsored virtual attendance)	2022	The John Maytham Show, CapeTalk (Interview)	2018
IBS ⁵ Early Career Biogeographers Conference	2021	<i>Harvesting Storm Water</i>	
iDiv ⁶ Joint Recruitment Symposium	2020	Pint of Science Cape Town	2018
SAAB-AMA-SASSB IVXV Joint Congress	2019	Biodiversity & Environmental Change in the CFR ⁷	2017

Research assistance

<i>Schoenus</i> (Cyperaceae) field assistance, collections & hydroponics for Dr T.L. Elliott & Prof A.M. Muasya (UCT)	2018
<i>Mimetes</i> & <i>Leucospermum</i> (Proteaceae) population monitoring for Dr J.A. Slingsby (UCT, SAEON)	2017
Legume symbiotic rhizobia collections & culturing for Dr M. Dlodlu (University of Swaziland)	2015